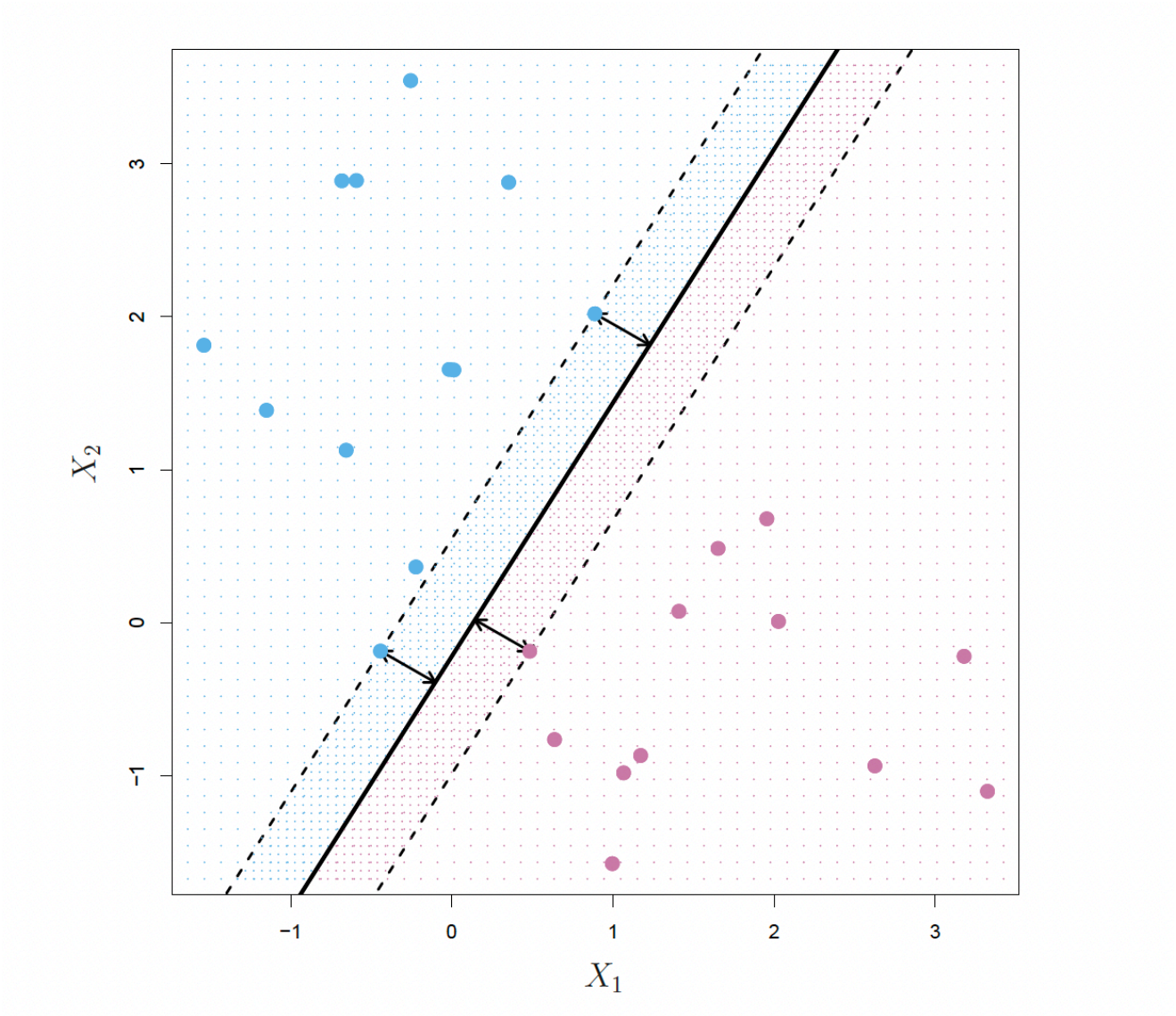
Maximal Margin Classifier = Optimal Separating hyperplane

Among all the separating hyperplanes, find the one that makes the biggest gap or margin between the two classes



Constrained Optimization Problem

$$\text{maximize } \mathrm{M}_{\beta_0,\beta_1,\ldots,\beta_p}$$

$$\text{subject to} \sum_{j=1}^{p} \beta_j^2 = 1$$

$$y_i(\beta_0 + \beta_1 x_{i1} + \ldots + \beta_p x_{ip}) \geq M$$

$$\text{For all} \quad i = 1, \ldots, N.$$